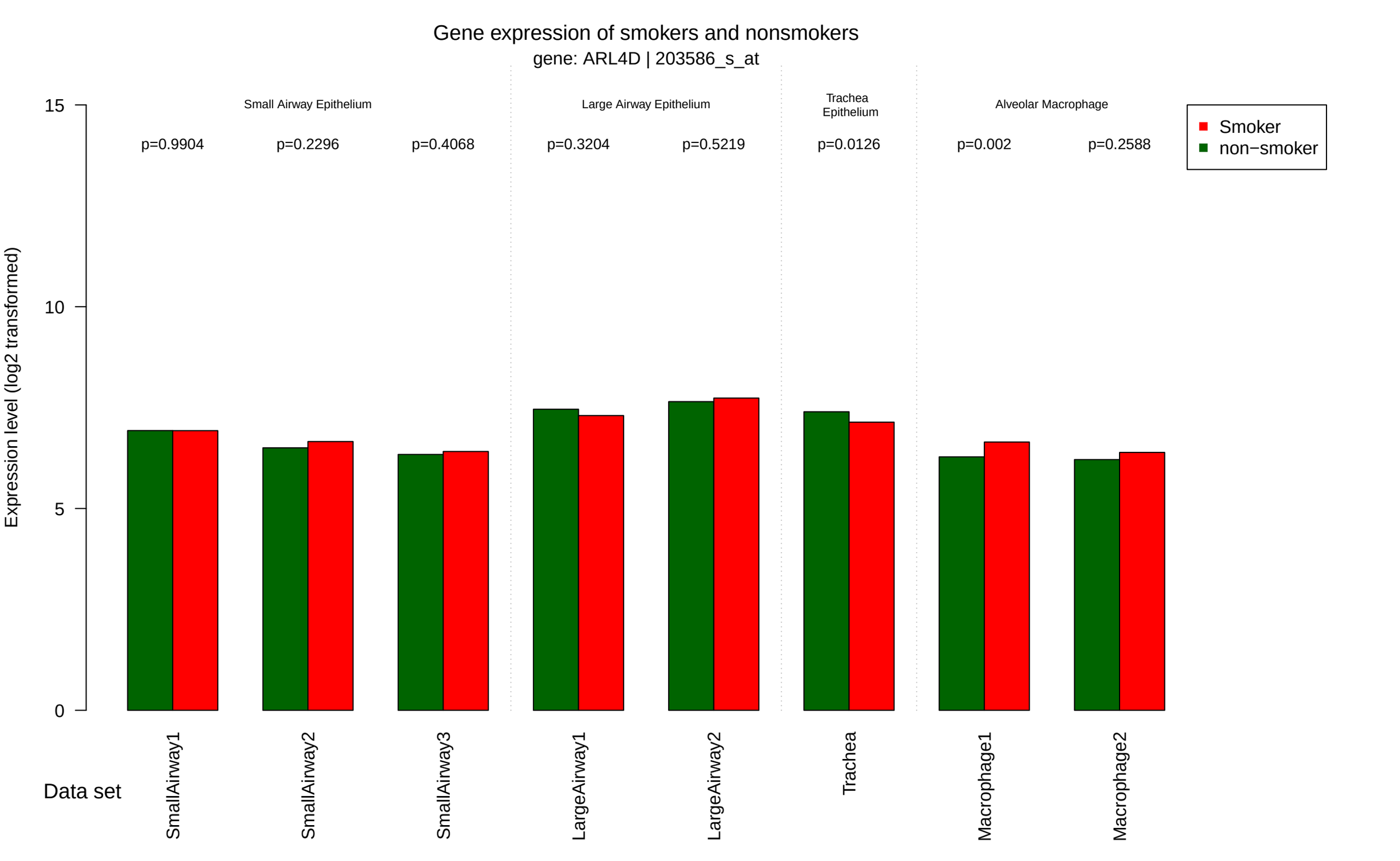
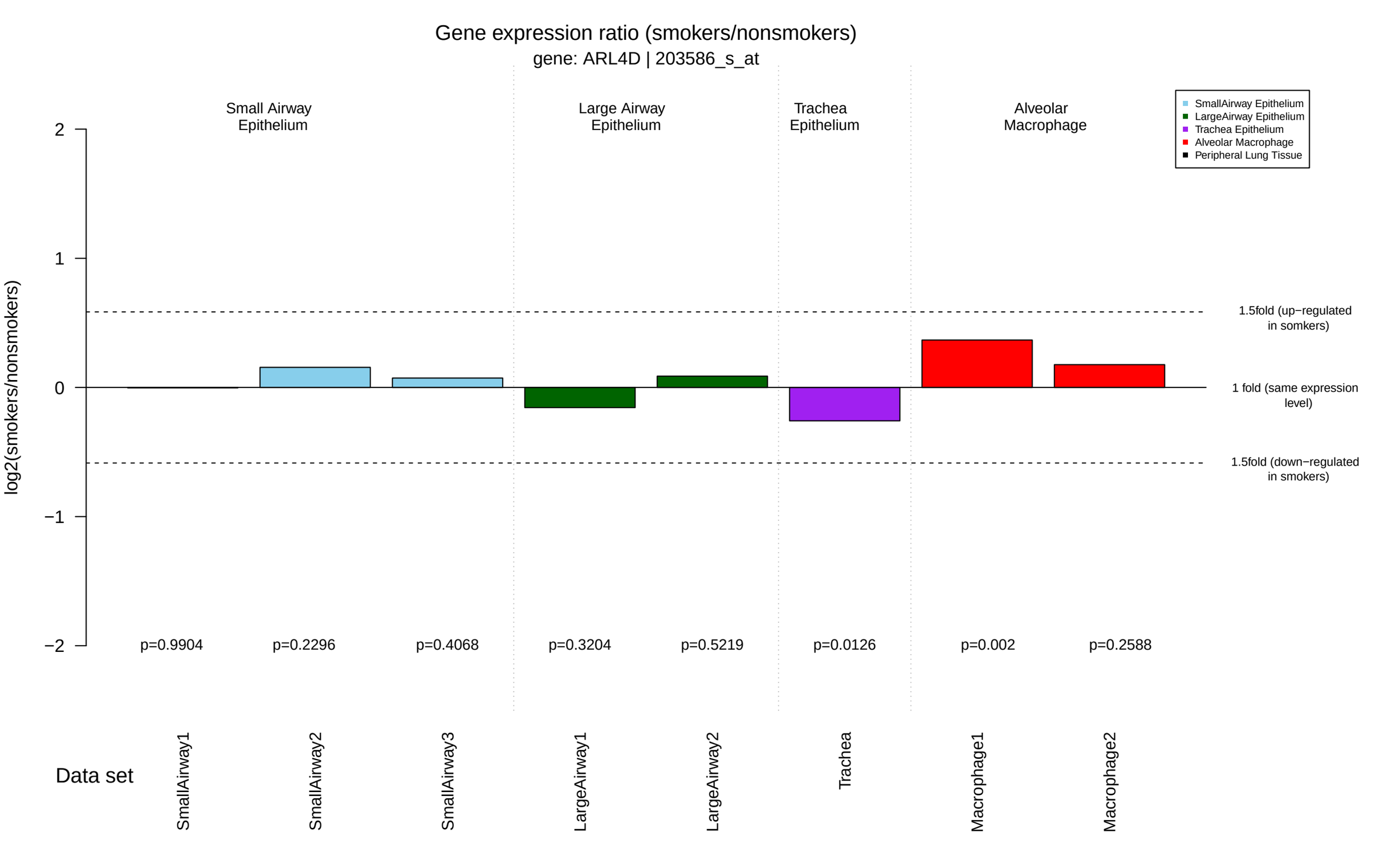
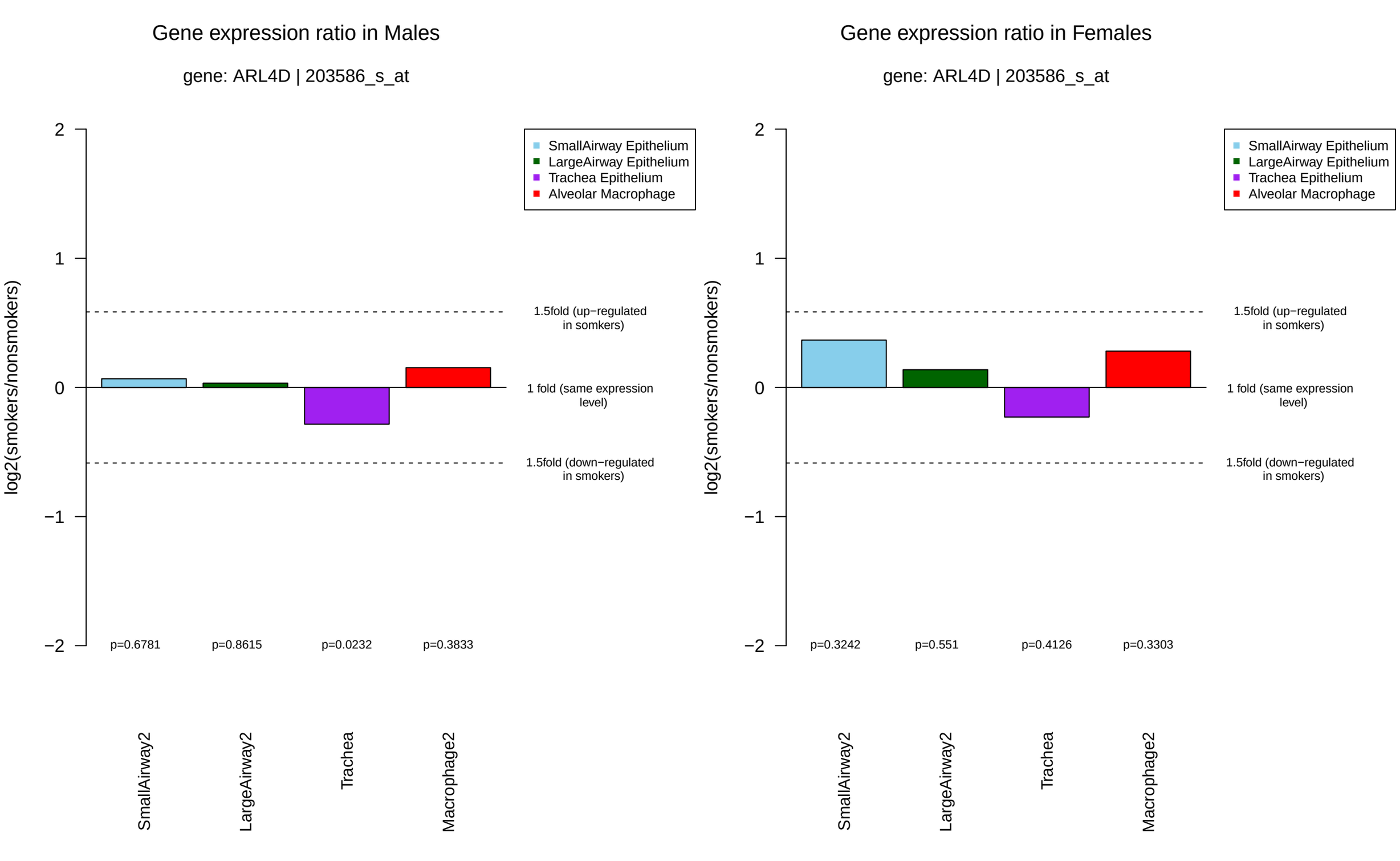
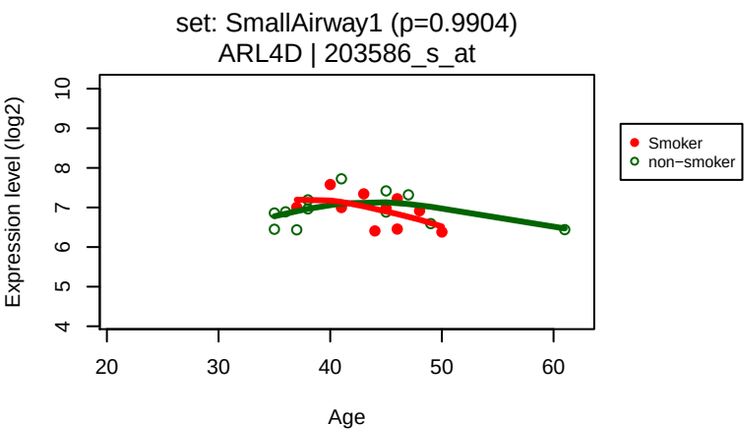
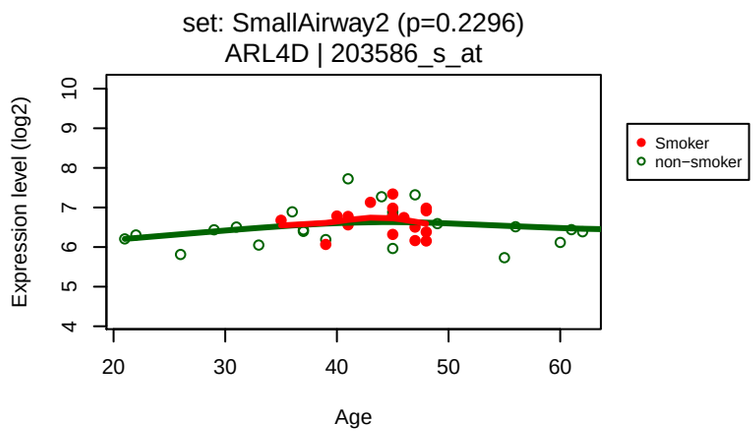
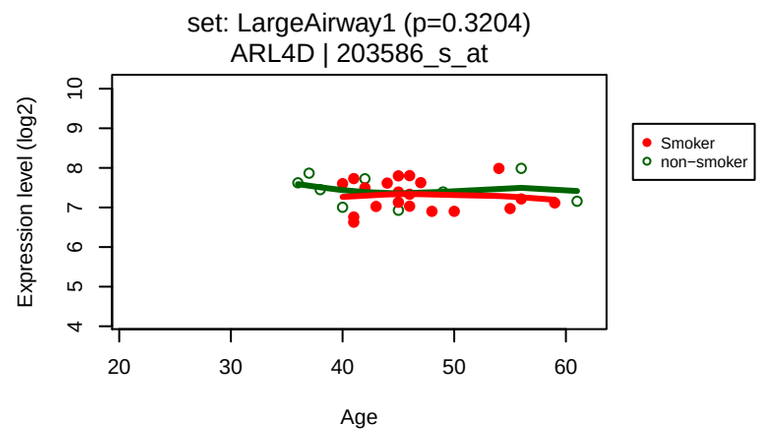

Gene expression of smokers and nonsmokers
gene: ARL4D | 203586_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9904
p=0.2296
p=0.4068
p=0.3204
p=0.5219
p=0.0126
p=0.002
p=0.2588
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

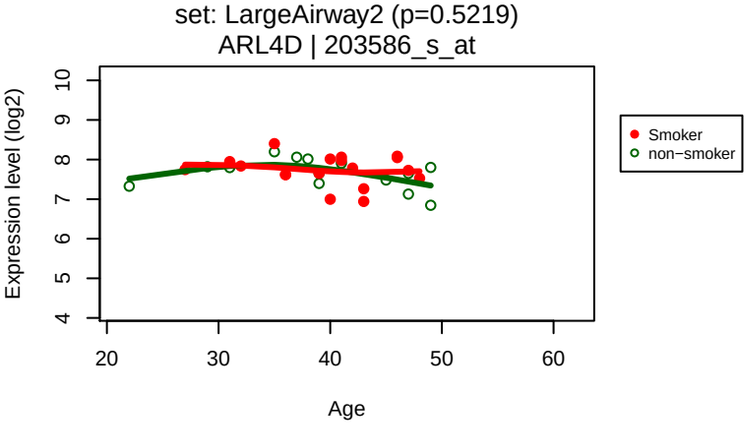

Gene expression ratio (smokers/nonsmokers)
gene: ARL4D | 203586_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9904
p=0.2296
p=0.4068
p=0.3204
p=0.5219
p=0.0126
p=0.002
p=0.2588
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

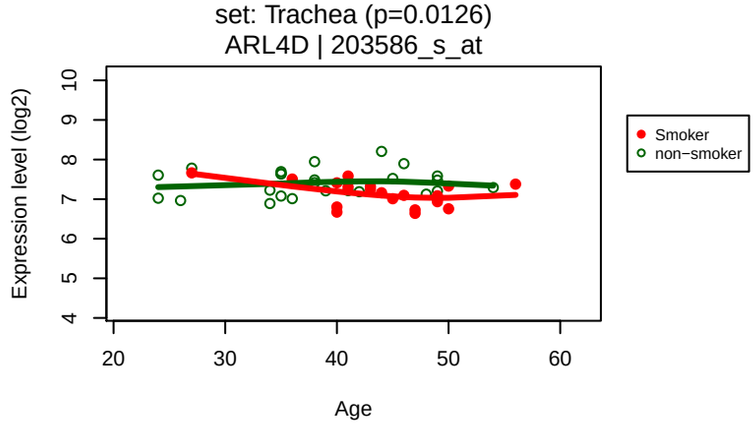

Gene expression ratio in Males
gene: ARL4D | 203586_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6781
p=0.8615
p=0.0232
p=0.3833
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARL4D | 203586_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3242
p=0.551
p=0.4126
p=0.3303
SmallAirway2
LargeAirway2
Trachea
Macrophage2

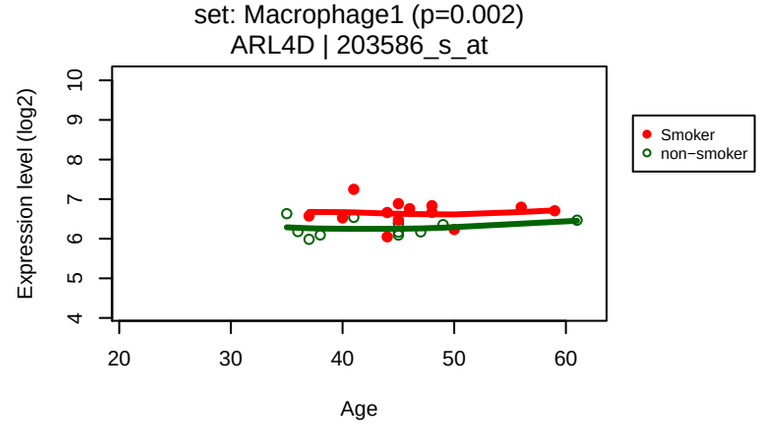

set: SmallAirway1 (p=0.9904)
ARL4D | 203586_s_at
Expression level (log2)
Age
Smoker
non−smoker

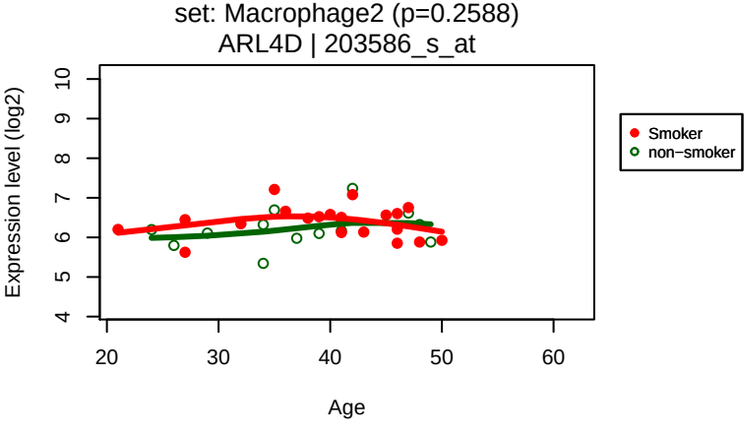

set: SmallAirway2 (p=0.2296)
ARL4D | 203586_s_at
Expression level (log2)
Age
Smoker
non−smoker

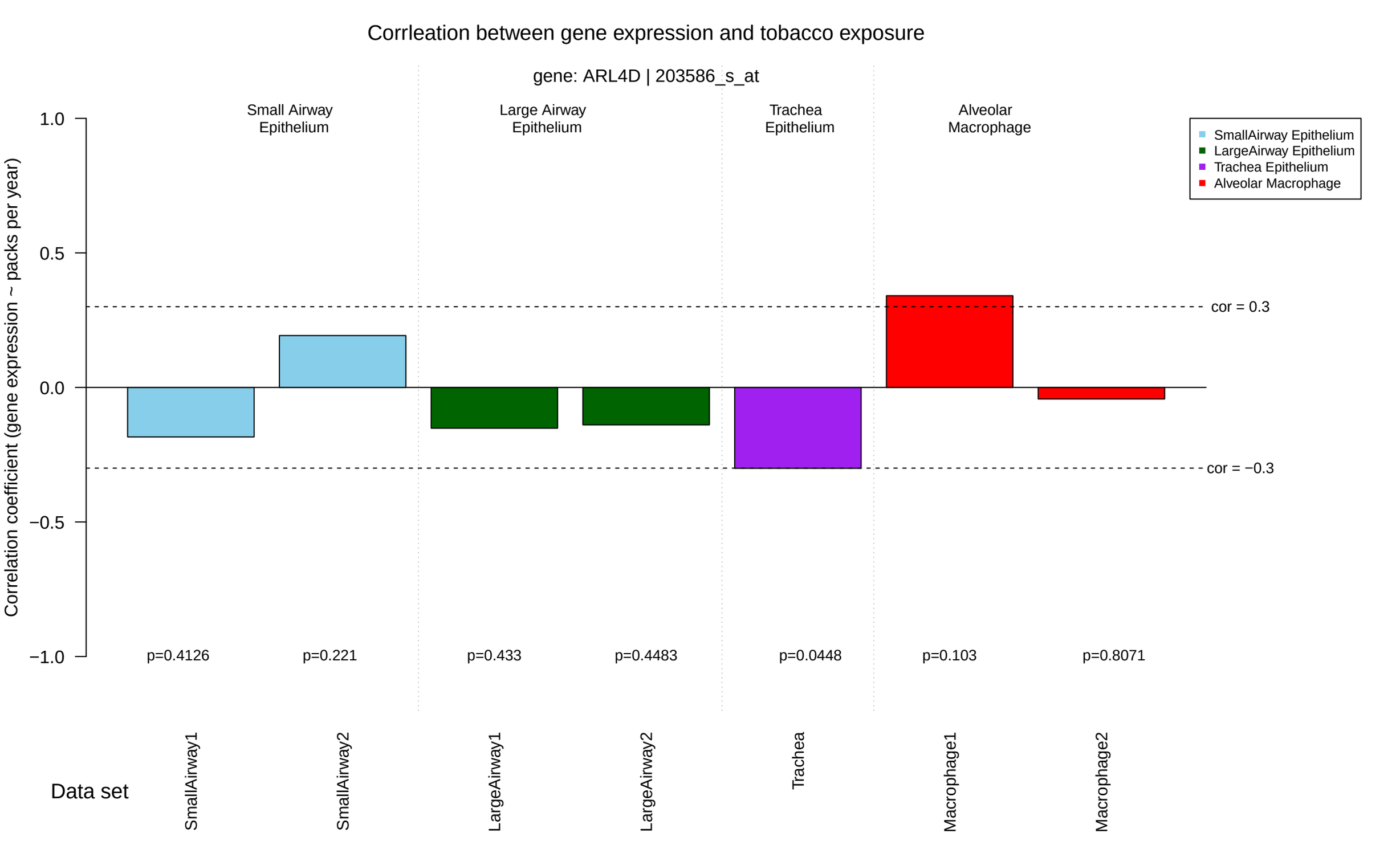

set: LargeAirway1 (p=0.3204)
ARL4D | 203586_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5219)
ARL4D | 203586_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0126)
ARL4D | 203586_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.002)
ARL4D | 203586_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2588)
ARL4D | 203586_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARL4D | 203586_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4126
p=0.221
p=0.433
p=0.4483
p=0.0448
p=0.103
p=0.8071
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2